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INTERNAL COMBUSTION ENGINE

Abstract

An internal combustion engine includes a blow-by gas passage. The internal combustion engine further includes a ventilation case provided at an intermediate position of the blow-by gas passage. The internal combustion engine further includes a positive crankcase ventilation (PCV) valve provided in the blow-by gas passage and attached to the ventilation case. The PCV valve regulates the amount of blow-by gas flowing through the blow-by gas passage. The internal combustion engine further a plate covering the PCV valve and attached to the ventilation case. The internal combustion engine further includes a cover covering the plate and in contact with part of the plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-035079 filed on Mar. 7, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to internal combustion engines.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2020-133607 (JP 2020-133607 A) discloses a positive crankcase ventilation (PCV) valve. The PCV valve is provided in a blow-by gas passage. The PCV valve regulates the amount of blow-by gas flowing through the blow-by gas passage. The blow-by gas passage is a passage for causing blow-by gas that has leaked from a combustion chamber of an internal combustion engine into a crankcase to flow into an intake system of the internal combustion engine. The PCV valve is covered by a plate in order to prevent the PCV valve from being removed from the internal combustion engine.

SUMMARY

[0004] The plate vibrates while the internal combustion engine is running. This may generate noise.

[0005] Hereinafter, means for solving the above issue and its functions and effects will be described.

[0006] One aspect of the present disclosure is an internal combustion engine. The internal combustion engine includes: [0007] a blow-by gas passage that causes blow-by gas that has leaked from a combustion chamber of the internal combustion engine into a crankcase to flow into an intake system of the internal combustion engine; [0008] a ventilation case provided at an intermediate position of the blow-by gas passage; [0009] a positive crankcase ventilation (PCV) valve provided in the blow-by gas passage and attached to the ventilation case, the PCV valve being configured to regulate an amount of the blow-by gas flowing through the blow-by gas passage; [0010] a plate covering the PCV valve and attached to the ventilation case; and [0011] a cover covering the plate and in contact with part of the plate.

[0012] The above configuration can reduce vibration of the plate while the internal combustion engine is running. As a result, noise can be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0014] FIG. 1 is a schematic diagram illustrating a configuration of an internal combustion engine according to an embodiment;

[0015] FIG. 2 is an exploded perspective view showing a plurality of members attached to a ventilation case of the internal combustion engine shown in FIG. 1; and

[0016] FIG. 3 is a cross-sectional view showing a part of the ventilation case of the internal combustion engine shown in FIG. 1 in which PCV valve is fixed.

DETAILED DESCRIPTION OF EMBODIMENTS

Configuration of the Internal Combustion Engine **100**

[0017] Hereinafter, an internal combustion engine according to an embodiment will be described with reference to the drawings. As shown in FIG. 1, the internal combustion engine **100** includes a

cylinder block **16**. The internal combustion engine **100** has a cylinder head **18** attached to an upper end portion of the cylinder block **16**. The internal combustion engine **100** has a crankcase **12** attached to a lower end portion of the cylinder block **16**. An upper portion of the cylinder head **18** is covered with a ventilation case **22**.

[0018] When the combustion of the fuel in the combustion chamber **10** of the internal combustion engine **100** is performed, blow-by gas leaks from the combustion chamber **10** of the internal combustion engine **100** to the crankcase **12**. The internal combustion engine **100** includes a blow-by gas reduction device for flowing blow-by gas leaked to the crankcase **12** to the intake system **14** of the internal combustion engine **100**. FIG. **1** shows an intake passage **14a** which is part of the intake system **14**.

[0019] The blow-by gas reduction device has a blow-by gas passage **20** for causing the blow-by gas to flow into the intake system **14** of the internal combustion engine **100**. The blow-by gas reduction device includes a positive crankcase ventilation (PCV) valve **24** for regulating the quantity of blow-by gas flowing through the blow-by gas passage **20**. The PCV valve **24** is provided in the blow-by gas passage **20** and is attached to the ventilation case **22**.

[0020] The blow-by gas passage **20** communicates with the crankcase **12**. The blow-by gas passage **20** extends in the cylinder block **16** and in the cylinder head **18**. The inside of the ventilation case **22** is part of the blow-by gas passage **20**. As described above, PCV valve **24** is attached to the ventilation case **22**. A hose **21** connects PCV valve **24** and the intake passage **14a**. The inside of the hose **21** is part of the blow-by gas passage **20**. In this way, the blow-by gas passage **20** communicates with the intake passage **14a**.

[0021] As described above, the ventilation case **22** is provided in the middle of the blow-by gas passage **20**. The PCV valve **24** is provided at an intermediate position of the blow-by gas passage **20**. As described above, PCV valve **24** regulates the amount of blow-by gas flowing through the blow-by gas passage **20**. Accordingly, the flow rate of the blow-by gas flowing from the crankcase **12** to the intake passage **14a** changes.

[0022] The internal combustion engine **100** includes a plate **26** covering the PCV valve **24** and attached to the ventilation case **22**. The internal combustion engine **100** includes a cover **28** covering the plate **26** and in contact with part of the plate **26**. The cover **28** is a urethane cover **28** made of urethane and surrounds PCV valve **24**.

How to Attach the PCV Valve **24**, the Plate **26**, and the Cover **28** to the Ventilation Case **22** in this Order

[0023] Referring to FIG. **2**, how to attach PCV valve **24**, the plate **26**, and the cover **28** to the ventilation case **22** in this order will be described. The ventilation case **22** has a mount portion **22a** that is a protruding portion of the ventilation case **22**. The mount portion **22a** has a mount hole **22b**.

[0024] First, in the first stage, the PCV valve **24** is attached to the ventilation case **22** in such a manner that the entire first portion **24a** of PCV valve **24** is located in the mount hole **22b** of the mount portion **22a**. The second portion **24b** of the PCV valve **24** that is continuous with the first portion **24a** is located outside the mount hole **22b**. The PCV valve **24** has the largest diameter in the second portion **24b**. Further, the second portion **24b** of PCV valve **24** covers the entire mount hole **22b**. Further, PCV valve **24** has a third portion **24c** extending from the second portion **24b** in a direction away from the mount hole **22b**. The third portion **24c** is a portion to which the hose **21** is connected.

[0025] Next, in a second stage, the plate **26** is attached to the ventilation case **22** over PCV valve **24**. More specifically, first, the third portion **24c** of PCV valve **24** is passed through a hole in the bent portion **26b** described later. The two cylindrical portions **22c** of the mount portion **22a** are then passed through the two holes in the cantilevered portion **26a** of the plate **26**. Then, the two cylindrical portions **22c** are deformed by being heated. As a result, the plate **26** is welded to the mount portion **22a**. Note that FIG. **3** shows the cylindrical portion **22c'** after the modification.

[0026] The plate **26** has a cantilevered portion **26a** partially fixed to the mount portion **22a** of the

ventilation case **22**, and a bent portion **26b** bent and extended from the cantilevered portion **26a**. The cantilevered portion **26a** is supported by two protrusions **22d** of the mount portion **22a**. The bent portion **26b** partially covers the second portion **24b** of the PCV valve **24** in the axial direction of the mount hole **22b**. The bent portion **26b** does not cover the third portion **24c** of the PCV valve **24** in the axial direction of the mount hole **22b**. The plate **26** further has a first side portion **26c** and a second side portion **26d** extending from the bent portion **26b** toward PCV valve **24** in the axial direction of the mount hole **22b**.

[0027] Finally, in a third stage, a cover **28** is attached to the ventilation case **22** over the plate **26**. PCV Valve **24** and its Surroundings

[0028] Referring to FIG. 3, PCV valve **24** and its periphery will be described. The cover **28** contacts the entire cantilevered portion **26a**. The second portion **24b** of the PCV valve **24** and the bent portion **26b** in the axial direction of the mount hole **22b** are separated from each other by a distance D. The second portion **24b** has a facing surface facing the bent portion **26b** and perpendicular to the axial direction of the mount hole **22b**. The distance D is a distance between the opposite surface of the second portion **24b** and the bent portion **26b** in the axial direction of the mount hole **22b**. The distance D is shorter than the length L of the first portion **24a** of PCV valve **24** in the axial direction of the mount hole **22b**.

Operation and Effect of the Present Embodiment

[0029] The internal combustion engine **100** includes a blow-by gas passage **20** for causing blow-by gas that has leaked from the combustion chamber **10** of the internal combustion engine **100** into the crankcase **12** to flow into the intake system **14** of the internal combustion engine **100**. The internal combustion engine **100** further includes a ventilation case **22** provided at an intermediate position of the blow-by gas passage **20**. The internal combustion engine **100** further includes a positive crankcase ventilation (PCV) valve **24** provided in the blow-by gas passage **20** and attached to the ventilation case **22**. The PCV valve **24** is configured to regulate the amount of blow-by gas flowing through the blow-by gas passage **20**. The internal combustion engine **100** further includes a plate **26** covering PCV valve **24** and attached to the ventilation case **22**. Internal combustion engine **100** further includes a cover **28** covering plate **26** and in contact with part of the plate **26**.

[0030] A cover **28** is in contact with part of the plate **26**. This configuration can reduce vibration of the plate **26** while the internal combustion engine **100** is running. As a result, noise can be suppressed.

[0031] Also, since the cover **28** is in contact with part of the plate **26**, this increases the resonant frequency of the plate **26**. As the resonance frequency moves away from the forced frequency, the vibration of the plate **26** is suppressed.

[0032] The ventilation case **22** has a mount portion **22a** that is a protruding portion of the ventilation case **22**. PCV valve **24** is attached to the ventilation case **22** as will be described below. The entire first portion **24a** of PCV valve **24** is inserted into the mount hole **22b** of the mount portion **22a**. Further, the second portion **24b** of PCV valve **24** that is continuous with the first portion **24a** is located outside the mount hole **22b**. Further, the second portion **24b** of PCV valve **24** covers the entire mount hole **22b**. The PCV valve **24** has a third portion **24c** extending from the second portion **24b** in a direction away from the mount hole **22b**. The PCV valve **24** has the largest diameter in the second portion **24b**.

[0033] The plate **26** has a cantilevered portion **26a** partially fixed to the mount portion **22a** of the ventilation case **22**, and a bent portion **26b** bent and extended from the cantilevered portion **26a**. The bent portion **26b** partially covers the second portion **24b** of the PCV valve **24** in the axial direction of the mount hole **22b**. The bent portion **26b** does not cover the third portion **24c** of the PCV valve **24** in the axial direction of the mount hole **22b**. The distance D between the second portion **24b** of the PCV valve **24** and the bent portion **26b** in the axial direction of the mount hole **22b** is smaller than the length L of the first portion **24a** of the PCV valve **24** in the axial direction of the mount hole **22b**.

[0034] In the above configuration, the bent portion **26b** partially covers the second portion **24b** of the PCV valve **24** in the axial direction of the mount hole **22b**. The distance D between the second portion **24b** of the PCV valve **24** and the bent portion **26b** in the axial direction of the mount hole **22b** is smaller than the length L of the first portion **24a** of the PCV valve **24** in the axial direction of the mount hole **22b**. Therefore, the bent portion **26b** restricts the first portion **24a** of PCV valve **24** from being removed from the mount hole **22b**. That is, if the plate **26** is not removed from the ventilation case **22**, it is difficult to remove the first portion **24a** of PCV valve **24** from the mount hole **22b**. Therefore, it is possible to restrain the PCV valve **24** from being removed from the ventilation case **22**.

[0035] The cover **28** contacts the entire cantilevered portion **26a**.

[0036] Since the cover **28** is in contact with the entire cantilevered portion **26a**, it is easier to suppress the oscillation of the plate **26** as compared with a configuration in which the cover **28** is in contact with only a part of the cantilevered portion **26a**.

[0037] The plate **26** is welded to the mount portion **22a**.

[0038] According to the above configuration, the plate **26** is less likely to be removed.

[0039] The cover **28** is a urethane cover **28** made of urethane and surrounds PCV valve **24**.

[0040] In the above-described configuration, the heat insulating power of PCV valve **24** by the cover **28** is increased.

Example of Change

[0041] The present embodiment can be modified and implemented as follows. The present embodiment and modification examples described below may be carried out in combination of each other within a technically consistent range.

[0042] In the above embodiment, the upper portion of the cylinder head **18** is covered by the ventilation case **22**. However, this is only an example. For example, the ventilation case **22** may be attached to the cylinder block **16**.

[0043] In the above embodiment, the plate **26** has a first side portion **26c** and a second side portion **26d**. At least one of the first side portion **26c** and the second side portion **26d** may be omitted.

[0044] In the above-described embodiment, the plate **26** is welded to the mount portion **22a**.

Alternatively, the plate **26** may be caulked to the mount portion **22a**.

[0045] In the above-described embodiment, as described above, the distance D is shorter than the length L. The distance D may be greater than or equal to the length L. In the above embodiment, the cover **28** contacts the entire cantilevered portion **26a**. However, the cover **28** may only touch a part of the cantilevered portion **26a**.

[0046] In the above embodiment, the cover **28** is a urethane cover **28** formed of urethane. However, this is only an example. For example, the cover **28** may be formed of a resin other than urethane.

Claims

1. An internal combustion engine, comprising: a blow-by gas passage that causes blow-by gas that has leaked from a combustion chamber of the internal combustion engine into a crankcase to flow into an intake system of the internal combustion engine; a ventilation case provided at an intermediate position of the blow-by gas passage; a positive crankcase ventilation valve provided in the blow-by gas passage and attached to the ventilation case, the positive crankcase ventilation valve being configured to regulate an amount of the blow-by gas flowing through the blow-by gas passage; a plate covering the positive crankcase ventilation valve and attached to the ventilation case; and a cover covering the plate and in contact with part of the plate.

2. The internal combustion engine according to claim 1, wherein the ventilation case includes a mount portion that is a protruding portion of the ventilation case, and the positive crankcase ventilation valve is attached to the ventilation case in such a manner that a first portion of the positive crankcase ventilation valve is entirely located in a mount hole of the mount portion, a

second portion of the positive crankcase ventilation valve that is continuous with the first portion is located outside the mount hole, and the second portion of the positive crankcase ventilation valve covers an entirety of the mount hole, the positive crankcase ventilation valve includes a third portion extending from the second portion in a direction away from the mount hole, and the positive crankcase ventilation valve has a largest diameter in the second portion, the plate includes a cantilevered portion partially fixed to the mount portion of the ventilation case and a bent portion bent from the cantilevered portion, the bent portion partially covers the second portion of the positive crankcase ventilation valve in an axial direction of the mount hole, and the bent portion does not cover the third portion of the positive crankcase ventilation valve in the axial direction of the mount hole, and a distance between the second portion of the positive crankcase ventilation valve and the bent portion in the axial direction of the mount hole is smaller than a length of the first portion of the positive crankcase ventilation valve in the axial direction of the mount hole.

3. The internal combustion engine according to claim 1, wherein the ventilation case includes a mount portion that is a protruding portion of the ventilation case, and the positive crankcase ventilation valve is attached to the ventilation case in such a manner that a first portion of the positive crankcase ventilation valve is entirely located in a mount hole of the mount portion, a second portion of the positive crankcase ventilation valve that is continuous with the first portion is located outside the mount hole, and the second portion of the positive crankcase ventilation valve covers an entirety of the mount hole, the plate includes a cantilevered portion partially fixed to the mount portion of the ventilation case and a bent portion bent from the cantilevered portion, and the bent portion partially covers the second portion of the positive crankcase ventilation valve in an axial direction of the mount hole, and the cover is in contact with an entirety of the cantilevered portion.

4. The internal combustion engine according to claim 2, wherein the plate is welded to the mount portion.

5. The internal combustion engine according to claim 1, wherein the cover is a urethane cover made of urethane and surrounds the positive crankcase ventilation valve.
